

Dr. Megha Babu

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Professional Profile

Early-career computational physicist and Earth-space interactions researcher with expertise in machine learning, deep learning, anomaly detection, signal processing, and time-series analysis applied to space, upper atmosphere, and geophysical observational datasets. PhD graduate in Space Science and Technology from the University of Trento and Fondazione Bruno Kessler (FBK), with doctoral research focused on applying deep learning in the field of Sun and Earth system studies. Experienced in scientific programming, statistical validation, uncertainty analysis, and reproducible data-analysis workflows for monitoring-oriented geophysical applications.

Research Interests

Ionospheric & Space Physics • Detection & Monitoring of Electromagnetic Signals • Machine Learning for space and Geophysical Observations • Remote Sensing • Signal Processing • Anomaly Detection • Statistical Analysis of Observational Data • Earth System Science • Scientific Computing

Education

PhD in Space Science and Technology (SST) 2022 – 2026

University of Trento & Fondazione Bruno Kessler (FBK), Italy
National Interest Doctorate (Dottorato di Interesse Nazionale)

- Research Title: Deep-learning-based anomaly detection in ionospheric observational data and statistical association with seismic activity.
- Supervisors: Dr. Marco Cristoforetti, Prof. Roberto Iuppa

Five Year Integrated Master of Science in Physics 2017 – 2022

Amrita Vishwa Vidyapeetham, India
First Class with Distinction – CGPA 9.10/10

Research Experience

Short-Term Scientific Mission (STSM)

COST Action CA24101 (FuSe) Sept 2026

National University of Science and Technology "Politehnica" Bucharest, Romania

Host: Assoc. Prof. Ciprian-Octavian Truičă

- Applying transformer-based (TimesNet, ViT) and diffusion-based (DDPM) deep learning architectures, fine-tuned via LoRA, to DEMETER ICE electric field data for seismo-ionospheric anomaly detection.
- Delivering a technical report and presentation to WG3 (Data Integration and Engineering) of COST Action CA24101, with outputs including an open-source repository and a planned journal manuscript.

PhD Researcher Nov 2022 – July 2026

University of Trento & Fondazione Bruno Kessler (FBK), Italy

- Developed reproducible machine-learning pipelines for anomaly detection in ionospheric and geophysical time-series data using satellite observations.

- Applied deep-learning models, including LSTM autoencoders, to investigate transient ionospheric perturbations and their statistical association with seismic activity.
- Performed end-to-end processing of observational datasets, including raw satellite data extraction, preprocessing, feature engineering, statistical validation, and uncertainty analysis.
- Contributed to interdisciplinary research integrating machine learning, space physics, and geophysical analysis, leading to peer-reviewed publications and scientific dissemination at multiple international conferences and workshops.

Research Internship – Ionospheric Research Group

Jan 2025 – Jun 2025

National Observatory of Athens, Greece

- Conducted collaborative research on ionospheric variability, using observational ground-based ionospheric datasets and scientific analysis workflows.
- Contributed to interdisciplinary, multi-instrumental research discussions on seismo-ionospheric phenomena.

Master Thesis Intern

2021 – 2022

Indian Institute of Technology (IIT) Indore, India

- Master's thesis: *Impacts of Coronal Mass Ejection on Earth's Magnetosphere* – investigated the geomagnetic and magnetospheric response to CME-driven structures.
- Contributed to the statistical analysis of ~ 290 corotating interaction regions (CIRs) and ~ 110 interplanetary sheaths identified in solar-wind and geomagnetic-index data spanning Solar Cycle 24 (2008–2019), comparing their solar-cycle variation and geoeffectiveness.
- Findings contributed to two peer-reviewed publications in *Solar Physics* (2022) and *Advances in Space Research* (2023) on the comparative characteristics and geoeffectiveness of interplanetary sheaths and CIRs.

Projects

- **Machine Learning for the Identification of Seismic Precursors through Ionospheric Anomaly Detection using satellite and ground-based observations**
Duration: Jan 2025 – Jun 2025 *Host Institute:* National Observatory of Athens (NOA), Greece, in collaboration with University of Trento & Fondazione Bruno Kessler (FBK), Italy
Role: Visiting Intern *Funding:* National Interest Doctorate Programme in Space Science and Technology (SST), Italy
- **Deep Learning for Detecting Time-Transient Phenomena in the Ionosphere and Correlation with Seismo-Induced Events**
Duration: Nov 2022 – July 2026 *Host Institute:* University of Trento, Fondazione Bruno Kessler (FBK) & INFN, Italy
Role: PhD Researcher *Funding:* National Interest Doctorate Programme (Dottorato di Interesse Nazionale) in SST, Italy
- **Impacts of Coronal Mass Ejections on Earth's Magnetosphere**
Duration: 2021 – 2022 *Host Institute:* Indian Institute of Technology (IIT) Indore, India in collaboration with Amrita Vishwa Vidyapeetham, Coimbatore, India.
Role: Master's Thesis Researcher *Supervisor:* Dr. Rajkumar Hajra, Dr. Naresh Raghava
- **Interplanetary Sheaths and Corotating Interaction Regions: A Comparative Statistical Study on Their Characteristics and Geoeffectiveness**
Duration: 2021 – 2022 *Host Institute:* Indian Institute of Technology (IIT) Indore, India
Role: Master's Thesis Researcher *Supervisor:* Dr. Rajkumar Hajra

- **A Comparative Study on Geoeffective and Non-Geoeffective Corotating Interaction Regions**
Duration: 2021 – 2022 *Host Institute:* Indian Institute of Technology (IIT) Indore, India
Role: Master's Thesis Researcher *Supervisor:* Dr. Rajkumar Hajra
- **Comparative Study of Pulsars Based on Emission Mechanism**
Duration: May-June 2021 *Host Institute:* Society for Space Education Research and Development (SSERD), Bangalore
Role: Student Researcher

Workshops & Scientific Activities

- Sigma Xi – The Scientific Research Honor Society, Member **2026 – Present**
- COST Action Working Group Member – *Testing Fundamental Physics with Seismology* (FuSe), Europe **2026 – Present**
- Member Researcher – Istituto Nazionale di Fisica Nucleare (INFN) - TIFPA **2025 – 2026**
- Industrial Problem Solving with Physics (IPSP), University of Trento, Italy **2025**
- European Space Weather Week (ESWW), Coimbra, Portugal **2024**
- PITHIA-NRF Training School, INGV Rome, Italy **2023**
- MaSAG23 Summer School and Conference, INGV Rome, Italy **2023**
- IUCAA Summer School in Astronomy and Astrophysics, Pune, India **2021**

Conference Presentations

1. **Deep Learning for the Detection of Ionosphere Electric Field Perturbations and Correlation with Seismo-Induced Events – Oral**
Testing Fundamental Physics with Seismology (FuSe) Annual Meeting **2026**
2. **Deep Learning Model for Detecting Global Ionospheric Electric Field Perturbations and Seismic Correlation – Oral**
M. Babu
EGU General Assembly 2026 **2026**
3. **Data-Driven Deep Learning Model for Identifying Ionospheric Electric Field Perturbations and Seismic Correlation – Poster**
American Geophysical Union (AGU) Fall Meeting **2025**
4. **Machine Learning for Detecting Time-Transient Phenomena in the Ionosphere and Correlation with Seismo-Induced Events – Oral**
M. Babu, M. Cristoforetti, R. Iuppa
EGU General Assembly 2025 **2025**
5. **Deep Learning Model for Identifying Seismic-Induced Ionospheric Electric Field Perturbations – Oral**
111th Congress of the Italian Physical Society (SIF), Palermo **2025**
6. **Machine Learning for Detecting Time-Transient Phenomena in the Ionosphere and Correlation with Seismo-Induced Events – Oral**
Electromagnetic Studies of Earthquakes and Volcanoes (EMSEV) Conference, Crete, Greece **2024**

Awards & Recognition

- **First Best Communication Award** – Geophysics and Atmospheric Physics, Italian Physical Society (SIF) **2025**
- **Young Researcher Award** – Electromagnetic Studies of Earthquakes and Volcanoes (EMSEV) Conference, Greece **2024**
- **Silver Medal** – Integrated MSc Physics Programme, Amrita Vishwa Vidyapeetham, India **2022**

Technical Skills

Programming & Scientific Computing: Python, Object-Oriented Programming (OOP), Git, Linux, Jupyter Notebook, Reproducible Scientific Workflows, LaTeX

Machine Learning & Quantitative Analysis: Machine Learning, Deep Learning, LSTM Autoencoders, CNN, RNN, Neural Networks, Anomaly Detection, Time-Series Analysis, Feature Engineering, Statistical Modelling, Uncertainty Analysis

Signal Processing & Data Analysis: Geophysical/Space Observational Data Analysis, Image Analysis, Multi-Instrument Data Processing, Data Preprocessing, Statistical Validation, Pattern Recognition

Libraries & Frameworks: NumPy, Pandas, SciPy, Scikit-learn, PyTorch, Matplotlib, Plotly, Seaborn, Origin, Qiskit

Research & Professional Skills: Technical Writing, Scientific Communication & Presentation, Interdisciplinary Collaboration, Independent Research, Problem-Solving, Leadership, Self-Learning

Languages

Malayalam (Native) • English (Full Professional Proficiency – publications, conferences, and doctoral research conducted in English) • Italian (Basic/Conversational)

Professional Service

Invited Peer Reviewer – *Earth Science Informatics* (Springer Nature) **2026**

Academic References

Dr. Marco Cristoforetti

Head, Data Science for Industry and Physics Unit
Fondazione Bruno Kessler (FBK), Italy
mcristofo@fbk.eu

Dr. Anna Belehaki

Research Director, Ionospheric Research Group
National Observatory of Athens, Greece
belehaki@noa.gr

Dr. Rajkumar Hajra

Professor (特任教授), School of Earth and Space Sciences
University of Science and Technology of China, China.
rhajra@ustc.edu.cn

Full Publication List

1. Hajra, R., Sunny, J. V., Babu, M., & Nair, A. G. (2022). Interplanetary sheaths and corotating interaction regions: A comparative statistical study on their characteristics and geoeffectiveness. *Solar Physics*, 297(7), 97.
DOI: 10.1007/s11207-022-02020-6
2. Sunny, J. V., Nair, A. G., Babu, M., & Hajra, R. (2023). A comparative study on geoeffective and non-geoeffective corotating interaction regions. *Advances in Space Research*, 71(1), 268–274.
DOI: 10.1016/j.asr.2022.09.051
3. Vitale, V., Neubüser, C., Battiston, R., Follega, F. M., Burger, W. J., Babu, M., et al. (2023). Electron signal induced by GRB 221009A on charged particle telescopes of POES and MetOp satellites. *The Astrophysical Journal*, 952(2), 159.
DOI: 10.3847/1538-4357/ace108
4. Amoroso, S., Babu, M., Bartocci, S., Battiston, R., Beolè, S., Burger, W. J., et al. (2025). Multispacecraft observations of the 27-day periodicity in galactic protons from 2018 to 2019. *The Astrophysical Journal*, 988(2), 181.
DOI: 10.3847/1538-4357/ade2e2
5. Perinelli, A., Follega, F. M., Babu, M., Bartocci, S., Battiston, R., Beolè, S., et al. (2025). In-depth analysis of selected major solar events with the HEPP-L particle detector onboard CSES-01 in low-Earth orbit. *Proceedings of Science*, PoS(ICRC2025), 1354.
6. Amoroso, S., Babu, M., Bartocci, S., Battiston, R., Beolè, S., Burger, W. J., et al. (2026). Measurement of the re-entrant all-electron spectrum by the High-Energy Particle Detector on board the China Seismo-Electromagnetic Satellite. *Astroparticle Physics*, 175, 103170.
DOI: 10.1016/j.astropartphys.2025.103170
7. Babu, M., Cristoforetti, M., Battiston, R., & Iuppa, R. (2026). Data-driven deep learning model for detecting ionospheric electric field perturbations and seismic correlation. *Remote Sensing*, 18(9), 1324.
DOI: 10.3390/rs18091324
8. Babu, M., Cristoforetti, M., & Iuppa, R. (2026). Deep learning model for identifying seismic-induced ionospheric electric field perturbations. *Il Nuovo Cimento C – Colloquia and Communications in Physics*, Article 143.
DOI: 10.1393/ncc/i2026-26143-y

Thesis Abstracts

PhD Thesis

Deep Learning Model for Identifying Ionospheric Electric Field Perturbations and Seismic Correlation
Handle: hdl.handle.net/20.500.14242/376840

Abstract – The quest for dependable earthquake-induced perturbations has driven investigations into the ionosphere. Recent advancements in satellite observations and machine learning offer novel opportunities to investigate these potential correlations via data-driven methodologies. This thesis examines ionospheric electric field disturbances through the application of deep learning methodologies to global space-based data. The research employs electric field measurements from the DEMETER satellite ICE mission, concentrating on very low frequency (VLF) electric field power spectral density data to examine ionospheric variability on a global scale. A deep learning framework using an LSTM autoencoder is developed to learn the underlying background behavior of the ionosphere and to detect anomalous perturbations in the data. This study utilised an adaptive rolling-window training strategy to address the highly non-stationary characteristics of geophysical datasets, continuously updating the model in

response to fluctuating background conditions. This methodological framework integrates deep learning and statistical validation to systematically examine potential correlations between identified ionospheric anomalies and seismic events. The findings indicate that deep learning models can effectively extract significant ionospheric electric field anomalies from extensive satellite datasets and detect statistically significant associations with following earthquakes under the given framework. Additionally, the study attempts a feasibility study on a multi-instrument approach by integrating ground-based ionosonde observations to examine potential coupling between disturbances in the lower and upper ionosphere. This thesis develops an adaptable and extensible deep learning framework for the detection of ionospheric perturbations from global space-based observations and to study the association with earthquakes. The proposed methodology is intended to be flexible for integration with alternative datasets and signatures supporting ongoing research into lithosphere–ionosphere interactions and enhancing the advancement of data-driven approaches for investigating potential seismo-ionosphere coupling.

Master's Thesis

Impacts of Coronal Mass Ejection on Earth's Magnetosphere

Abstract – Space weather events are initiated on the Sun and can cause immense impacts on the Earth's magnetosphere-ionosphere-atmosphere system. The events cause geomagnetic storms which can damage the satellite communication, navigation and power grid systems. Two such events are Corotating Interaction Regions (CIRs) and Coronal Mass Ejections (CMEs). These are the major sources of southward component of Interplanetary Magnetic Fields, which are responsible for geomagnetic storms. The interplanetary remnant of coronal mass ejection is Interplanetary Coronal Mass Ejection (ICME). The dark void of ICME with a smooth rotation of interplanetary magnetic field is Magnetic Cloud (MC). Sheaths are the turbulent region of plasma and magnetic field formed between ICME and shock. This study explores the long-term solar-wind measurements upstream of the Earth during Solar Cycle 24, from January 2008 to December 2019, to compare the characteristic features, geoeffectiveness and energy variations of Coronal Mass Ejections. Earth was found to be encountered by 164 geomagnetic storms during this period. Among these 88 storms were caused by CIRs and 76 by sheaths and ICMEs. The occurrence of geoeffective storms follows the F10.7 solar flux variations. A quantitative study considering 20 geoeffective CME events on the energetics of the solar wind-magnetosphere-ionosphere system during CME events for Solar Cycle 24 is presented. Both sheath and MC were evaluated separately. For 20 considered events sheath efficiency (the ratio of solar wind energy input into the magnetosphere to the total solar wind energy), varies from a minimum value of 0.05 to a maximum value of 0.43 and MC efficiency varies from a minimum value of 0.07 to a maximum value of 0.75. The average energy efficiency was found to be ~ 0.19 for sheath and ~ 0.27 for MC. We can conclude that the mean energy efficiency of MC is more than that of the sheath. The typical characteristic solar-wind, geomagnetic activity parameters and energetics given in this article can be useful for modelling and prediction purposes.